MODEL QUESTION PAPER

CHEMICAL ENGINEERING THERMODYNAMICS

Max.Marks : 75

Part A

I. Answer all questions in one word or one sentence. Each question carries 1 mark

9 X 1 = 9 Marks

1. ……………….. properties of a system do not depend on the quantity of matter contained in it.
2. ……….. and ……….. arepath functions.
3. State the first law of thermodynamics.
4. Define isothermal process.
5. Define entropy.
6. Indicate*one* reference property.
7. Mach number is the ratio of ……………….
8. Define order of reaction.
9. Define space time.

Part B

II. Answer any EIGHT questions from the following.Each question carries 3 marks

8 X 3 = 24 Marks

1. Determine the number of degrees of freedom for a system containing water and water vapour in equilibrium.
2. Drawthe schematic representation of heat engine and explain its principle.
3. Explainthe concept of saturation pressure of a substance.
4. Discuss onthe limitations of first law of thermodynamics.
5. From a reservoir at 600 K, 1000 J of heat is transferred to an engine that operates on the Carnot cycle. The engine rejects heat to a reservoir at 300 K. Determine the thermal efficiency of the cycle and the work done by the engine.
6. State the second law of thermodynamics.
7. Explain the concept of fugacity.
8. Describe COP of a refrigerator.
9. Differentiate single and multiple reactions.
10. Describe elementary reactions.

Part C

Answer ALL questions. Each question carries 7 marks

6 X 7 = 42 Marks

III. Differentiate homogeneous and heterogeneous systems with examples.

OR

IV. A system consisting of some fluid is stirred in a tank. The rate of work done on the system by the stirrer is 2.25 hp. The heat generated due to stirring is dissipated to the surroundings. If the heat transferred to the surroundings is 3400 kJ/h, determine the change in internal energy.

V. Prove that Cp -Cv = R for an ideal gas.

OR

VI. Describeheat of reaction and standard heat of reaction.

VII. Sketch P-V diagram for Carnot cycle and explain the processes involved.

OR

VIII. State Hess's law of constant heat summation and calculate the heat of formation of methane gas from the following heat of combustion data:

$CH_4\,(g) + 2O_2(g) \longrightarrow CO_2\,(g) + 2H_2O\,(l)\ ;\ \Delta H^{\circ}_{298} = -890.94$ kJ

$C\,(s) + O_2\,(g) \longrightarrow CO_2\,(g)\ ;\Delta H^{\circ}_{298} = -393.78$ kJ

$H_2\,(g) + \frac{1}{2}\,O_2\,(g) \longrightarrow H_2O\,(l)\ ;\Delta H^{\circ}_{298} = -286.03$ kJ

IX. List and explain the desirable properties of a refrigerant.

OR

X. Describe the processes involved in air – standard Otto cycle with the help of T-S diagram.

XI. Describe Rankine cycle with the help of T-S diagram.

OR

XII. Describe throttling process.

XIII. Describe the principle and applications of MFR.

OR

XIV. List and explain the variables affecting rate of a reaction.

Prepared By:

Devi P

Scrutinized By:

Surya Sathyan

Course :Chemical Engineering Thermodynamics

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Mark Distribution

Module	Hr/Module	Marks / Module	Type of Questions							
			Part A		Part B		Part C		Total	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
1	9	25	3	3	2	6	2	14	7	23
2	16	43	2	2	4	12	4	28	10	42
3	12	33	2	2	2	6	4	28	8	36
4	8	22	2	2	2	6	2	14	6	22
Total	45	123	9	9	10	30	12	84	31	123

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Cognitive Level Mark Distribution

Cognitive Level	Marks	% of Marks
Remembering	35	28
Understanding	71	58
Applying	17	14
Analysing	-	
Evaluating	-	
Creating	-	
Total	**123**	**100**

Question Wise Analysis

Course :Chemical Engineering Thermodynamics

Q.No	Module Outcome	Cognitive Level	Marks	Time
I.1	M 1.02	R	1	
I.2	M 1.02	R	1	
I.3	M 1.05	R	1	
I.4	M 2.02	R	1	
I.5	M 2.06	R	1	
I.6	M 3.04	R	1	
I.7	M 3.01	R	1	
I.8	M 4.02	R	1	
I.9	M 4.04	R	1	
II.1	M 1.03	U	3	
II.2	M 1.04	U	3	
II.3	M 2.01	U	3	
II.4	M 2.05	R	3	
II.5	M 2.06	A	3	
II.6	M 2.06	R	3	
II.7	M 3.04	U	3	
II.8	M 3.02	U	3	
II.9	M 4.02	R	3	
II.10	M 4.02	R	3	

III.	M 1.01	U	7	
IV.	M 1.05	A	7	
V.	M 2.02	U	7	
VI.	M 2.03	R	7	
VII.	M 2.06	U	7	
VIII.	M 2.04	A	7	
IX.	M 3.02	R	7	
X.	M 3.03	U	7	
XI.	M 3.03	U	7	
XII.	M 3.01	U	7	
XIII.	M 4.03	U	7	
XIV.	M 4.02	U	7	